Short-mode white noise in a radiation universe: spatial curvature and the measured temperature

Opus 5.5, Meatbag Operator

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Verdict

Two short sound waves of nearly equal wavelength beat against each other. Their stress is quadratic, so it has a slow component at the beat wavenumber K, however long that beat may be. This report asks what the slow component does. The answer is largely the story of one trace-free tensor built up by that stress, $S_{ij} \to W(x)\,\mathrm{TF}(\hat{q}\hat{q})$. The volume variable ζ cannot see it, the curvature $\mathcal{R}$ reads it along $\hat{K}$, and the sky reads its change in time along $\hat{n}$.

Spatial curvature: yes. Short-wavelength modes alone give the comoving curvature perturbation a finite white-noise term, $P_{\mathcal{R}}(K) \to \epsilon^4 A_0(t) > 0$, with no $1/K^2$ enhancement and no suppression by powers of K. The coefficient is explicit, Eqs. (7)–(9). Its direction-dependent part is the projection of S_{ij}, and it grows as $\ln(qt)$ once the short modes are inside the horizon. The answer depends on the threading in the definition of $\mathcal{R}$: the conserved ζ has no response at all, to all orders.

Measured temperature: yes, but not as curvature, and the white noise alone is not the signal. The full normalised map has a nonzero K^0 quadrupole, Eq. (13). Yet a finite $K \to 0$ limit is generic for any observer-local quantity (Prop. 2), so what distinguishes this result is the generated part: $-\frac{1}{2}\hat{n}^i\hat{n}^j\Delta S_{ij}$, the time *difference* of the tensor that appears in $\mathcal{R}$, growing as $-8\ln(t_o/t_\star)$. Direct short-mode terms of the same order (emission surface, observer boost, ray) add no logarithm within about ± 0.1; this is numerical, not proved. They still matter for the power: the generated term dominates $\mathcal{Q}_2^2$ for $qt_\star \lesssim 10$, but above that the emission term, which has zero mean and rms growing as $qt_\star$, takes over (Table 2). An exact nonlinear calculation agrees with the map to better than 10^{-4} out to $(x_\star, x_o) = (5, 160)$ at $K = 0$, and converges to the same limit at $K \neq 0$.

Of the three possible outcomes, this is the first: an explicit nonzero coefficient. A white-noise $P(K)$ does not imply a flat C_ℓ, and no C_ℓ is computed here.

1 Setup and the definition of $\mathcal{R}$

We use the main report's conventions, in which the fluid supplies its own coordinates. For an irrotational radiation fluid, $Tu_\mu = -\mathcal{T}\partial_\mu t$ defines a potential t, and the X^i are comoving material labels. In these coordinates

$$g = N^2(-\mathrm{d}t^2 + \gamma_{ij}\mathrm{d}X^i\mathrm{d}X^j), \qquad N = a_1 t e^\nu = \mathcal{T}/T, \qquad \gamma = e^{2\zeta}e^H, \quad \mathrm{tr}\,H = 0. \tag{1}$$

So ζ carries the volume of γ, H its shape, and ν the local clock. The primordial state is the regular solution with $\zeta = \epsilon\zeta_1(X)$ exactly and ζ_1 Gaussian, with $\gamma \to e^{2\zeta}\delta$ and $\nu, H \to 0$ as $t \to 0$. The spectrum is smooth with support in $q_{\min} < q < q_{\max}$.

What we mean by $\mathcal{R}$. A curvature perturbation needs two choices. *Slicing:* level sets of t, the hypersurfaces orthogonal to u. *Threading:* on each slice, the chart x in which the induced metric takes the form

$$h_{ij} = a^2 e^{2\mathcal{R}}(e^\chi)_{ij}, \qquad \mathrm{tr}\,\chi = 0, \quad \partial_j\chi_{ij} = 0, \qquad a = a_1 t. \tag{2}$$

Geometrically, $\mathcal{R}$ is the conformal factor of the comoving slice in the chart where the unimodular part is transverse. That is what makes it a curvature: χ does not enter the scalar curvature linearly, and $a^2\,{}^{(3)}R = -4\nabla^2\mathcal{R} + 8\mathcal{R}\nabla^2\mathcal{R} - 2(\partial\mathcal{R})^2 + \dots$. The price is that $\mathcal{R}$ is a nonlocal functional of the 3-geometry; the chart exists perturbatively and is unique at nonzero wavenumber. With $X = x + \xi(x,t)$ and $J = \det \partial X/\partial x$, taking the determinant gives the exact relation

$$\mathcal{R}(x,t) = \zeta(X) + \nu(X,t) + \tfrac{1}{3}\ln J. \tag{3}$$

As $t \to 0$, $\mathcal{R} = \zeta$ exactly. Through second order, with $\xi_1 = \partial\beta_1$, $m = \partial\partial\beta_1$, $\psi = \zeta + \nu$, and first-order threading fixed by $\chi_1 = 0$,

$$\mathcal{R}_2 = \psi_2 + \xi_1\cdot\partial\psi_1 - \tfrac{1}{6}\operatorname{tr} m^2 - \tfrac{1}{4}\hat{K}_i\hat{K}_j S_{ij}, \qquad S = H_2 + \xi_1\cdot\partial H_1 - \mathrm{TF}\, m^2. \tag{4}$$

Here our tensor makes its first appearance. This form is verified off shell, for generic momenta and an arbitrary trace-free H_2, by matching the scalar curvature in the two charts (`check_chart_R.py`).

At first order, $x = qt/\sqrt{3}$, $\nu_1 = \cos x - j_0(x)$, $j_0 = \sin x/x$, and $H_1 = h(x)\,\mathrm{TF}(\hat{q}\hat{q})\zeta_1$ with $h = 6(1 + \cos x - 2j_0)$. This gives $\mathcal{R}_1 = j_0(x)\zeta_1$, which starts at 1 and decays inside the horizon.

2 First route: the decisive calculation

Take the seed $\zeta_1 = A\cos(q\cdot x) + B\cos[(q-K)\cdot x]$. The output at K is $(AB/2)\cos(K\cdot x)\,K_2$, where $K_2 = 2F_2$ is the coefficient of $z_1 z_2$ for the pair $(q, K-q)$ with $z_1 z_2 = AB/4$.

Why the limit is finite and direction dependent. One observation does most of the work. The trace-free equation has an entire Green function in the Cartesian components of K (main report, "Projector-free Cartesian solution"), so H_2 and ν_2 are continuous at $K = 0$: $H_2 \to a(x)\,\mathrm{TF}(\hat{q}\hat{q})$ and $\nu_2 \to n_2(x)$. No pole can arise from them. But the projector in (4) remains, and $\hat{K}\hat{K}\cdot\mathrm{TF}(\hat{q}\hat{q}) = \mu^2 - \frac{1}{3}$, with $\mu = \hat{K}\cdot\hat{q}$. So the direction is not averaged away, and the limit is not analytic in K. A constant survives, and it remembers the angle between the beat and the waves.

The $K = 0$ pair fields in closed form. All that remains is to evaluate two functions of time, and the limit makes this easy: with ζ along one axis the exact equations reduce to 1+1 dimensions. The pair sources integrate by hand, and all Si terms cancel:

$$n_2 = -\cos 2x - 6\cos x + 6j_0 + 3\frac{\sin 2x}{x} - 5j_0^2, \tag{5}$$

$$a = -78 - 36\cos x - 6\cos 2x + 72j_0 + 36\frac{\sin 2x}{x} - 24j_0^2 + 24\,\mathrm{Cin}(2x). \tag{6}$$

Substitution into (T), (N), (M) and (Ham) gives exact zeros, and mutated coefficients fail. Also $a = 2x^2 + O(x^4)$, $n_2 = \frac{5}{3}x^2 + O(x^4)$, which match the exact gradient data. Assembling (4), the terms linear in $\cos x$ cancel and we arrive at the main result for the curvature:

$$\boxed{F_2^{\mathcal{R}}(q, K-q) \xrightarrow{K\to 0} \tfrac{1}{2}I(x) - \tfrac{1}{8}\Big(\mu^2 - \tfrac{1}{3}\Big)W(x), \qquad \begin{aligned} I &= 4 + \cos^2 x - 5j_0^2, \\ W &= 24\,\mathrm{Cin}(2x) + 6\cos^2 x - 54 + 48j_0^2. \end{aligned}} \tag{7}$$

So the response is K^0: no pole and no vanishing constant. Writing $F_2 \to \alpha + \beta P_2(\mu)$ with $\alpha = I/2$ and $\beta = -W/12$,

$$\alpha = \frac{x^2}{3},\quad \beta = -\frac{x^2}{6} \quad (x \ll 1); \qquad \alpha \to 2 + \tfrac{1}{2}\cos^2 x,\quad \beta \to -2\big[\ln 2x + \gamma_E - \tfrac{9}{4} + \tfrac{1}{4}\cos^2 x\big] \quad (x \gg 1). \tag{8}$$

This is valid for $K \ll q$ and $Kt \ll 1$ at fixed t, including $qt \gg 1$.

The power spectrum. One might fear that the P_2 part dies under the integral, since its average over $\hat{q}$ vanishes. But power involves $|F_2|^2$, whose average does not. With $\langle P_2^2\rangle = \frac{1}{5}$,

$$\boxed{P_{22}(K\to 0) = A_0(t) = \frac{1}{\pi^2}\int_0^\infty q^2 \mathrm{d}q\, P_{\zeta_1}^2(q)\left[\alpha^2 + \frac{\beta^2}{5}\right] > 0.} \tag{9}$$

Early on, $A_0 \to \frac{7t^4}{540\pi^2}\int q^6 P^2 \mathrm{d}q$. Late, the bracket is dominated by $\frac{4}{5}\ln^2(qt)$. The quantity A_0 is dimensional power: the dimensionless spectrum is $K^3 A_0/2\pi^2$. Although the kernel is non-analytic in $\hat{K}$, the spectrum is isotropic, so P_{22} is constant as $K \to 0$. The exact kernels at finite K converge to (7) at all tested angles, including $|q_1| \neq |q_2|$. A direct quadrature for a smooth compact bump centred at q_0 with half-width $0.3q_0$ gives $P_{22}(K)/A_0 = 0.8589,\ 0.9565,\ 0.9867$ at $K/q_0 = 0.2,\ 0.1,\ 0.05$ for $q_0 t/\sqrt{3} = 1$, and $0.7685,\ 0.9327,\ 0.9782$ for $q_0 t/\sqrt{3} = 4$. These approach 1; the deficit shrinks by a factor of about 3.3 per halving of K, and we do not claim the exact power of the correction.

Why P_{13} is absent. This one is simple. $P_{13}(K) = 2\,\mathrm{Re}\langle X_1(K) X_3^*(K)\rangle$, and $X_1(K) \propto \zeta_1(K) = 0$ for $K < q_{\min}$, so it vanishes identically, as does P_{12}. This does not carry over to C_ℓ: $a_{\ell m}^{(1)}$ projects band modes onto ℓ through Bessel functions at finite distance, so $C_\ell^{(13)} \neq 0$ (main report).

Why the usual suppression fails. The logarithm is the memory of the short modes' effective stress (derived in Section 3). In the usual language $\dot{\mathcal{R}} \sim K^2 \Pi_S$, which looks suppressed. But the scalar stress potential obeys $\Pi_S \sim \hat{K}\hat{K}\Pi/K^2$, so the factor K^2 cancels. Nor is this scalar piece on its own: a single fixed tensor has helicity power $1:2:2$ (scalar:vector:tensor) over directions $\hat{K}$. The scalar white noise is therefore the same phenomenon as the known K^3 tail of induced gravitational waves [1, 2]. As an independent check, on the shell the $K \to 0$ limit of the Kohri–Terada form in the main report satisfies $F_T \to a/2$, agreeing to 3×10^{-11} for x up to 15.

Where each contribution comes from.

Primordial variable	None: $\mathcal{R} = \zeta$ at $t \to 0$, $\zeta_2 \equiv 0$. A local $\zeta \to \zeta + f\zeta^2$ would add a constant; here $f = 0$ by construction.
Constraint-required data at t_i	$F_2(x_i) = x_i^2/3 - \frac{1}{4}x_i^2(\mu^2 - \frac{1}{3}) + O(x_i^4)$. It is part of the same regular solution, vanishes as $t_i \to 0$, and is not an independent mode.
Measurement definition (threading)	Local products $h(1+\nu_1) - h^2/12 - \frac{1}{8}(\mu^2 - \frac{1}{3})h^2$ in K_2. They are bounded and contain the $\cos x$ terms that cancel against n_2 and a.
Evolution	n_2 (bounded, zero-mean oscillation at late times) and $-\frac{1}{4}(\mu^2 - \frac{1}{3})a$ (memory integral of the stress, with the logarithm).

A caution about threading. In material threading, the volume variable $\zeta + \nu$ has only n_2 at $K \to 0$: isotropic, local, with no logarithm. The isotropic part α of (7) can be removed by a local field redefinition. The $\hat{K}$-dependent part β cannot, since it is non-analytic in K, but it does vanish under a change of threading. So $\mathcal{R}$ alone does not establish a physical signal. For that we need the shear, and the sky.

3 Second route: what can be said exactly

Before turning to the sky, we ask how much of this survives without expanding in ϵ.

Proposition 1 (All orders, for smooth irrotational flow)**.**

(a) Conservation. $\partial_t \det\gamma = 0$, *so* $\zeta_K(t) = \zeta_K(0)$ *for all* K. *Physically,* $e^{3\zeta}$ *is the entropy per unit label. Below the band* $\zeta_K = 0$ *exactly, with no singular term, no constant and no power of* K.

(b) Relation to $\mathcal{R}$. *Eq.* (3). *Thus (a) does not constrain* $\mathcal{R} - \zeta$: ν, *the composition with* X, *and* $\ln J$ *all contribute at* K.

(c) Transversality. *Define* $\mathcal{S}$ *by* $\ln(J^{-2/3}(\partial X)^T e^{H\circ X}\partial X) = \mathcal{S} + 2\,\mathrm{sym}\,\partial\xi - \frac{2}{3}\delta\,\partial\cdot\xi$. *Then* $\partial_i\partial_j\chi_{ij} = 0$ *gives, for* $K \neq 0$, $(\partial\cdot\xi)_K = -\frac{3}{4}\hat{K}_i\hat{K}_j\mathcal{S}_{ij,K}$. *Therefore*

$$\mathcal{R}_K = \mathcal{C}_K - \tfrac{1}{4}\hat{K}_i\hat{K}_j\mathcal{S}_{ij,K}, \qquad \mathcal{C} = \zeta\circ X + \nu\circ X + \tfrac{1}{3}(\ln J - \partial\cdot\xi). \tag{10}$$

(d) Shear balance, retaining all short gradients. *Since* $\partial_t\sqrt{\gamma} = 0$, *equation (T) gives*

$$\partial_t\big(N^2\sqrt{\gamma}\,S^i{}_j\big) = -N^2\sqrt{\gamma}\;\mathrm{TF}\left[R^i{}_j[\gamma] - 2D^iD_j\nu + 2D^i\nu D_j\nu\right], \quad S^i{}_j = \tfrac{1}{2}\gamma^{ik}\partial_t\gamma_{kj}. \tag{11}$$

The right side has a nonzero long-wavelength component made from short modes: this is the effective stress.

(e) Relation to the observable. *The exact redshift (main report) is* $T_{\rm obs} = T_{\rm fluid}(o)\,e^{-I}$, $I = \int\sigma_{ij}e^ie^j\mathrm{d}t$. *With* $\hat{e} = e^{\zeta}e$, *the integrand is* $\frac{1}{2}(\partial_t e^H)_{ij}\hat{e}^i\hat{e}^j$, *where* $\hat{e}^Te^H\hat{e} = 1$. *By (a) it has no trace part, and* ζ *does not appear in it explicitly. So* ζ *reaches* Θ *only through the ray path and as a source of* H *and* ν, *and* ν *only through the emission and observation endpoints.*

What this proves, and what it does not. Only (a) is a genuinely all-orders constraint on the response, and it concerns ζ: a complete vanishing, stronger than any power of K. The other parts locate the response without fixing it. Part (c) is algebra on the chart definition and (d) is equation (T), so the constant cannot be obtained from them alone. What they do give is this. If $\mathcal{S}$ and $\mathcal{C}$ have bounded spectra at small K, (10) excludes a singular enhancement. It does *not* exclude a constant; it identifies the constant as the spectrum of $\mathcal{C}$ plus the $\hat{K}\hat{K}$ projection of $\mathcal{S}$. Stronger suppression is false, because the $O(\epsilon^2)$ value of that constant is the nonzero (7), with $\mathcal{S}_2 = W\,\mathrm{TF}(\hat{q}\hat{q})$, the same W. The boundedness hypothesis is checked only for the leading $O(\epsilon^4)$ spectrum. Beyond that, (c) makes $\xi_K \sim 1/K$ at second order, and this re-enters $\mathcal{S}$ at higher order. We expect the resulting convolutions to be finite but have not computed them. Parts (a), (b), (d), (e) are exact for any smooth solution; (c) also needs the transverse chart to exist. No gradient expansion of the short modes and no soft input is used.

An independent value for the logarithm. This route does earn one number on its own, and it is the most physical one. The $K \to 0$ mode of (d) needs only first-order fields as input. At $O(\epsilon^2)$ it reads $a'' + 2a'/t = -S_T$ (primes in t, $q = 1$), with

$$S_T = -12\cos x - 8\cos 2x + \frac{24\sin 2x}{x} - \frac{48\sin^2 x}{x^2} + \frac{16}{x^2} - \frac{16\sin 2x}{x^3} + \frac{16\sin^2 x}{x^4},$$

whose late-time mean is $-8/x^2$. Identifying $a'' + 2a'/t = (6/t^2)\Pi/\bar{\rho}$ gives $\Pi_{ij}/\bar{\rho} = 4\,\mathrm{TF}(\hat{q}\hat{q})$. This is simply the acoustic stress $\frac{4}{3}\bar{\rho}\langle v_iv_j\rangle$ with $v = \sqrt{3}\sin x\,\hat{q}\,\zeta_1$, computed from the first-order sound wave alone, and it yields $a \simeq 24\ln x$: a factor 6 from gravity and a factor 4 from the sound. So the coefficient of the logarithm in W, and hence the $-8 = -\frac{1}{3}\cdot 24$ in $\mathcal{Q}_2$, is fixed independently of the closed forms. The bounded terms are not.

Order of limits and nonuniformity. The calculation takes the $O(\epsilon^2)$ kernel first (so the $O(\epsilon^4)$ spectrum); then $K \to 0$ at fixed t; then, optionally, $qt \to \infty$. The opposite order, $K \to 0$ at fixed ϵ and then $\epsilon \to 0$, would need higher-order spectra to stay bounded uniformly as $K \to 0$. For Θ each order is finite (Prop. 2), though uniformity in ϵ is not shown. For $\mathcal{R}$ even that is the unchecked hypothesis above. So the exchange of $K \to 0$ and $\epsilon \to 0$ is an assumption. Two further nonuniformities matter. (i) Fixed K with $t \to \infty$ leaves the regime $Kt \ll 1$, and the logarithm then saturates into propagating modes, which this calculation does not treat. (ii) Fixed ϵ with $qt \to \infty$ fails through acoustic steepening; a simple-wave estimate gives $\epsilon x_s \approx \frac{1}{2}$ for $\zeta = 2\epsilon\cos qy$. Exact plane-symmetric evolution agrees: the $K = 0$ average satisfies $\langle f\rangle/\epsilon^2 - a \approx 0.39$ ($\epsilon = 0.01$, $x = 10$), 3.39 ($\epsilon = 0.03$, $x = 10$), scaling as ϵ^2, and spectral steepening appears at $\epsilon x \approx 0.3$–0.4. After a shock, (a) fails with entropy production. The results therefore need $\epsilon qt \ll 1$.

4 The measured temperature

Now we ask the question in the form an observer can answer. Prop. 1(e) simplifies matters at once: $T_{\rm fluid}(o)$ cancels on normalisation and only shear can produce anisotropy, so ν_2 and α enter only at higher order or not at all. The observer is the fluid line at fixed label X_o, with fixed proper time after $T = T_I$, and Θ is the normalised map, with the dipole removed at the end. Fields are Fourier analysed in X_o, since the observer is a label. In the chart x an extra bounded term $\xi_1 \cdot \partial\Theta_1$ would appear.

At $K \to 0$ the second-order map becomes a function of $\hat{n} \cdot \hat{q}$ alone, even in $\hat{n}$, with no dipole. For the seed,

$$\Theta(\hat{n}, X_o) \supset \frac{AB}{2} \cos(K \cdot X_o) \sum_{\ell \geq 2 \text{ even}} \mathcal{Q}_\ell \, P_\ell(\hat{n} \cdot \hat{q}). \tag{12}$$

The quadrupole coefficient is exactly

$$\boxed{\mathcal{Q}_2 = -\tfrac{1}{3}\big[W\big]_{x_\star}^{x_o} + \mathcal{Q}_2^{\rm em} + \mathcal{Q}_2^{\rm prod} + \mathcal{Q}_2^{\rm clk} + \mathcal{Q}_2^{\rm rem},} \tag{13}$$

where each piece has a plain meaning, with $\mathcal{Q}_2$ denoting the P_2 projection throughout.

- *Generated:* $-\frac{1}{2}\hat{n}^i\hat{n}^j[S_{ij}]_\star^o$, with $S = W\,\mathrm{TF}(\hat{q}\hat{q})$, the same tensor as in $\mathcal{R}$. This split is not unique: using W rather than the evolved a moves bounded threading terms out of the remainder, and $-\frac{1}{3}[a]$ alone would carry a ± 12 oscillation. Only the coefficient of the logarithm is independent of the choice.
- *Emission surface* (closed form): $\mathcal{Q}_2^{\rm em} = -\frac{2}{3} t_\star h'(t_\star)\nu_1(t_\star) \to 2x_\star \sin 2x_\star$ for $x_\star \gg 1$. The surface is displaced at first order and the local shear acts on that displacement, so this is a direct short-mode term.
- *Normalised products* of first-order maps, $\mathsf{P}_0(J_a J_b)$: for large separation, $\to 2(\sin^2 x_o + \sin^2 x_\star)$, the squared Doppler terms at both ends.
- *Clock:* $-D(\partial_o J_a + \partial_o J_b)$, with $D = [-t_I^2\nu_1(t_I) - \int_{t_I}^{t_o} t\nu_1 \mathrm{d}t]/t_o$.
- *Remainder:* the ray deflection and direction terms $X_1 \cdot \partial H_1'$, $H_1'(\hat{n}, v_1)$, plus $\frac{1}{6}[h^2]_\star^o$. These are double time integrals, given explicitly in `map_k0.py`.

Outside the horizon. For $qt \ll 1$ everything is local, and

$$\mathcal{Q}_2 = -\tfrac{2}{9} q^2 (t_o^2 - t_\star^2) + O(q^4 t^4), \tag{14}$$

which is the $z_1 z_2$ part of the local quadrupole $-\frac{1}{6}\Delta(t^2)\, e^{-2\zeta}\,\mathrm{TF}[\partial\partial\zeta - \partial\zeta\partial\zeta]_{nn}$. It is a nonzero K^0 response, not suppressed relative to K^2 the way a linear long adiabatic mode is.

Inside the horizon. Table 1 gives the evaluated pieces with $t_I = t_\star/2$. The large $\cos x_o$ oscillation in a is absent once the same combination W is used. For $x_o \gg x_\star \gg 1$,

$$\mathcal{Q}_2 \simeq -8 \ln \frac{x_o}{x_\star} + 4 \sin^2 x_o + 2x_\star \sin 2x_\star + O(1). \tag{15}$$

Three things are visible here: the stress memory, the observer's boost and the emission surface. The coefficient -8 of the generated term is exact. The claim that the other terms add no logarithm is numerical. Fitting $\mathcal{Q}_2 + \frac{1}{3}[W] - \mathcal{Q}_2^{\rm em}$ over log-spaced $x_o \in (30, 1000)$ gives a logarithm coefficient of -0.03 ± 0.02, 0.00 ± 0.12, $+0.09 \pm 0.11$ for $x_\star = 2, 5, 8$ (bootstrap standard deviations of the fit), at most about 2% of the generated -8. A regression of $\mathcal{Q}_2$ itself gives a $\cos 2x_o$ coefficient near the predicted -2, the observer's second-order boost. We would expect this: subhorizon potentials decay as x^{-2}, so ray rectification has no $1/t$ rate, and first-order $\sigma_{nn} \propto \mu^2 - \frac{1}{3}$ vanishes on the sound-cone resonance $\mu^2 = \frac{1}{3}$. Neither observation is a proof.

$x_\star$	x_o	$\mathcal{Q}_2$	$-\frac{1}{3}[W]$	rem	em	prod	clk
0.1	0.2	−0.019594	−0.0201	+0.0003	−0.0000	+0.0000	+0.0002
0.5	1	−0.319767	−0.5230	+0.1519	−0.0251	+0.0122	+0.0642
1	3	−4.268883	−2.7998	+1.3472	−0.2881	+0.1195	−2.6478
3	10	+0.292937	−8.1918	+0.1877	+6.8473	+0.8380	+0.6116
5	20	−15.3640	−10.1915	−1.8555	−7.3100	+3.4488	+0.5442
5	40	−20.7210	−16.5168	+1.7639	−7.3100	+1.5628	−0.2209
5	80	−24.8166	−21.0832	+0.3781	−7.3100	+3.0030	+0.1955
5	160	−33.9877	−28.5273	+0.9593	−7.3100	+0.9047	−0.0144

Table 1: The $K \to 0$ quadrupole coefficient in (13), $t_I = t_\star/2$. Printed values are evaluated from the closed fields and the report's map, by quadrature.

Testing it without the map. A formula this intricate deserves a test that does not use it. We solved the exact plane-symmetric Einstein–radiation system with $\zeta = 2\epsilon \cos qy$, integrated exact null rays with physical emission roots, the proper-time clock and the exact sky normalisation, and averaged over observer labels. No perturbative map formula enters. Richardson extrapolation in ϵ reproduces $\mathcal{Q}_2$ with differences 3×10^{-8} to 5×10^{-6} for seven time pairs up to $(x_\star, x_o) = (5, 20)$, and at most 8×10^{-5} in the logarithmic regime, for $(5, 40), (2, 40), (5, 80), (2, 80), (5, 160)$. Omitting any single piece of the map is detected.

The same solver also tests $K \neq 0$ directly, with $\zeta = 2\epsilon[\cos(ny) + \cos((n-1)y)]$ in units $K = 1$. This is our seed with $\hat{q} \parallel \hat{K}$, and the K channel is read from the observer-label harmonic $\cos y_0$. With times set by the mean q, the difference $\mathcal{Q}_2(K) - \mathcal{Q}_2$ falls as K^2 (Table 3). So the $K \to 0$ limit of the $K \neq 0$ channel is the $K = 0$ value, as continuity requires. Other angles between $\hat{q}$ and $\hat{K}$ are not tested forward. The limit does not depend on $\hat{K}$ because Θ has no projector, but this rests on Prop. 2.

The spatial spectrum of the temperature. Writing the $\ell = 2$ part as $Q_{ij}\hat{n}^i\hat{n}^j$,

$$P_{Q_{ij}Q_{ij}}(K \to 0) = \frac{3\epsilon^4}{8\pi^2} \int q^2 \mathrm{d}q \, P_{\zeta_1}^2(q) \, \mathcal{Q}_2^2(qt_\star, qt_o, qt_I) > 0. \tag{16}$$

This is white noise in the observer-label field of the measured quadrupole, not an angular spectrum. It says that quadrupoles measured by distant observers have a K^0 spectrum, and C_2 receives this region only with phase-space weight $\propto \int K^2 \mathrm{d}K$.

How impressed should we be by that constant? Not very, on its own, for the following reason.

Proposition 2 (Locality fixes the form, not the coefficient)**.** *Order by order in ϵ, for compact spectra, the kernels of ν_r, H_r and Θ are entire in the Cartesian momenta, and $\Theta(\hat{n}, X_o)$ depends only on the past light cone of the observer. Hence every $P_\Theta^{(r,s)}(K)$ is continuous at $K = 0$: there is no pole, and a constant is the generic outcome.*

Sketch. First-order kernels are entire in q, since h is even with $h(0) = 0$. The induction of the main report's Cartesian proposition extends this to ν_r, H_r. The map consists of finite time integrals of these with phases $e^{ik\cdot\hat{n}\lambda}$, plus polynomial root, clock and normalisation terms. Wick contractions then integrate over compact sets. This does not apply to $\mathcal{R}$, which has the $\hat{K}\hat{K}$ projector and ξ.

So a K^0 spectrum in Θ does not by itself show a generated long mode; the squared Doppler term has one too. The content is in the coefficient. Since P_{QQ} depends on $\mathcal{Q}_2^2$, the zero mean of the emission term gives no protection: this is the same point as $\langle F_2 \rangle$ against $\langle |F_2|^2 \rangle$. The generated term dominates at moderate $x_\star$ or large $x_o/x_\star$, and the direct emission term dominates once $x_\star \gg \ln(x_o/x_\star)$. At $x_o/x_\star = 4$ the emission share of $\langle \mathcal{Q}_2^2 \rangle$ is 0.42 at $x_\star = 5$ and 0.96 at $x_\star = 20$, while the generated share falls from 1.06 to 0.14 (Table 2).

What is and is not curvature here. We can now close the circle. A constant-in-time long S_{ij} is a coordinate rescaling and is invisible in Θ, which sees only ΔS. That removal of the exactly homogeneous, time-independent part does not make the connected spectrum vanish, because the stress makes S_{ij} time dependent. The adiabatic expectation $\Theta_2 \sim K^2 t^2 \mathcal{R}$ fails because this $\mathcal{R}$ is not an adiabatic long mode: it arrives with an order-K^0 shear. In the end $\mathcal{R}$ and Θ read the same tensor, via $\hat{K}\hat{K}$ and $\hat{n}\hat{n}$ respectively.

A Supporting tables

$x_\star$	$x_o/x_\star$	rms $\mathcal{Q}_2$	rms gen	rms em	$\langle \mathrm{gen}^2\rangle/\langle\mathcal{Q}_2^2\rangle$	$\langle \mathrm{em}^2\rangle/\langle\mathcal{Q}_2^2\rangle$
1	4	3.12	3.66	0.34	1.38	0.01
3	4	4.49	9.80	6.82	4.77	2.31
5	4	10.60	10.89	6.86	1.06	0.42
10	4	16.08	11.06	14.12	0.47	0.77
20	4	29.81	11.12	29.22	0.14	0.96
5	32	24.83	27.43	6.86	1.22	0.08

Table 2: Division of the temperature white-noise power, with q averaged over $\pm 30\%$ with flat weight at fixed $t_o/t_\star$ and $t_I = t_\star/2$. Fractions need not sum to one because of cross terms.

K/q (n)	1/7.5 (8)	1/15.5 (16)	1/31.5 (32)	1/63.5 (64)
$(x_\star, x_o) = (1, 3)$	+0.294	+0.0697	+0.0169	+0.0042
$(3, 10)$	–	+0.249	+0.0672	+0.0170
$(5, 20)$	–	–	+2.77	+0.706

Table 3: Forward test at $K \neq 0$: $\mathcal{Q}_2(K) - \mathcal{Q}_2$ for the collinear pair, Richardson extrapolated in ϵ.

B Status of the claims

Claim	Status
ζ conservation, identities (a)–(e)	Exact derivation
a, n_2, I, W, $F_2^{\mathcal{R}}$ limit, (4)	Exact symbolic check, negative controls
$K \to 0$ continuity of kernels	Proved (entire Green function); numerically confirmed
$\mathcal{Q}_2$ evaluation and all pieces	Exact formula; quadrature; forward test at $K = 0$ up to $x_o = 160$
$\mathcal{Q}_2(K) \to \mathcal{Q}_2$ at $K \neq 0$, $\hat{q} \parallel \hat{K}$	Forward test; K^2 convergence
Finite $K \to 0$ limit of Θ at each order (Prop. 2)	Proof sketch; forward check at second order
Absence of further logarithms in $\mathcal{Q}_2$; form (15)	Numerical bound, $\lesssim 2\%$ of the coefficient; no proof
Exchange of $K \to 0$ and $\epsilon \to 0$; bounded $\mathcal{R}$ spectrum beyond $O(\epsilon^4)$	Assumed, not shown
Long-time behaviour beyond $\epsilon q t \ll 1$; shocks	Not resolved; outside the smooth finite-time setup of the question
Forward test with $\hat{q}$ at a generic angle to $\hat{K}$	Not performed (needs 3+1 evolution); $\mathcal{R}$ kernels at generic angle are checked by the Green solution
Complete C_ℓ from this white noise; superhorizon-to-subhorizon entry of the output	Not computed; the question does not require them

C Code and reproduction

All code is in `solution/followup/`; `run_followup.sh` runs everything (about 40 minutes, peak memory about 350 MB).

File	Purpose
`plane_reduction.py`, `k0_fields.py`	reduced equations from the Ricci formula; closed-form proofs
`check_chart_R.py`	off-shell scalar-curvature invariance of (4)
`check_followup.py`	negative controls, tensor limit, stress, helicity, omission tests
`map_k0.py`	complete $\mathcal{Q}_2$ and pieces, Table 1
`forward_plane.py`, `forward_log.py`	exact PDE and exact rays against $\mathcal{Q}_2$, including the logarithmic regime
`forward_K.py`, `k_series.py`	the same at $K \neq 0$; exact t-series of the K-channel initial data
`finite_K.py`	finite-K limit and $P_{22}(K)/A_0$
`late_time.py`, `late_log_bound.py`	regression for (15); bound on a further logarithm
`power_split.py`, `nonuniform.py`	Table 2; amplitude nonuniformity

The forward solver takes f' at nonzero wavenumber from the momentum constraint: evolving (T) directly excites the $-K^2/3$ scalar instability excluded by the constraints. At small K the initial data must also be accurate in the K channel, because (N) has the constraint-violating homogeneous solution $\nu \propto t^2$. With gradient data alone, an $O(x_s^4)$ error at the start time x_s grew into a spurious $K \to 0$ offset of about -0.44 at $(3, 10)$. The series through t^8 removes it.

A stand-alone check of the decisive fields (`minimal_check.py`), which needs only SymPy and prints zero residuals:

```
"""Stand-alone: the K->0 pair fields solve the plane-symmetric equations (q=1, x=t/sqrt3)."""

# fmt: off
import sympy as sp

t, y, e, z1, z2 = sp.symbols("t y e z1 z2", positive=True)
x = t / sp.sqrt(3)
j0, c = sp.sin(x) / x, sp.cos(x)
C2 = sp.Function("C2")(t)                       # Cin(2x); C2'(t) = (1-cos 2x)/t
n1, h1 = c - j0, 6 * (1 + c - 2 * j0)
n2 = -sp.cos(2 * x) - 6 * c + 6 * j0 + 3 * sp.sin(2 * x) / x - 5 * j0**2
a2 = -78 - 36 * c - 6 * sp.cos(2 * x) + 72 * j0 + 36 * sp.sin(2 * x) / x - 24 * j0**2 + 24 * C2
g = z1 * sp.exp(sp.I * y) + z2 * sp.exp(-sp.I * y)
Z = e * g
f = e * h1 * g + e**2 * a2 * z1 * z2             # H = f diag(2/3,-1/3,-1/3); gamma = e^{2Z} exp H
nu = e * n1 * g + e**2 * n2 * z1 * z2
D = lambda u, v: sp.diff(u, v).subs(sp.Derivative(C2, t), (1 - sp.cos(2 * x)) / t)
fy, ny, zy = D(f, y), D(nu, y), D(Z, y)
ex = sp.exp(-2 * f / 3 - 2 * Z)
Ff = ex * (-fy**2 / 18 + fy * ny / 3 + fy * zy / 6 + D(fy, y) / 6 + 2 * ny**2
           + 4 * ny * zy - 2 * D(ny, y) + zy**2 - D(zy, y))
ft, nt = D(f, t), D(nu, t)
lap = ex * (D(ny, y) - (zy + fy / 3) * ny + 2 * (zy - fy / 6) * ny)
T_ = D(ft, t) + 2 * (1 / t + nt) * ft + 2 * Ff                                  # (T)
N_ = (3 * D(nt, t) - lap - 2 * ex * ny**2 + ft**2 / 6
      + 3 / t**2 * (sp.exp(-2 * nu) - 1))                                       # (N)
for name, eq in (("T", T_), ("N", N_)):
    first = sp.simplify(sp.diff(eq, e).subs(e, 0))
    second = sp.expand(sp.diff(eq, e, 2).subs(e, 0) / 2).coeff(z1, 1).coeff(z2, 1)
    print(name, "first:", first, "pair:", sp.simplify(sp.expand(sp.expand_trig(second))))
```

References

[1] K. Kohri and T. Terada, Phys. Rev. D 97 (2018) 123532.

[2] R.-G. Cai, S. Pi and M. Sasaki, Phys. Rev. D 102 (2020) 083528.